4/10

CHE 313 Transport Phenomena III

Quiz #3 (11/11/25) **Name:** DiataRodrisuCt

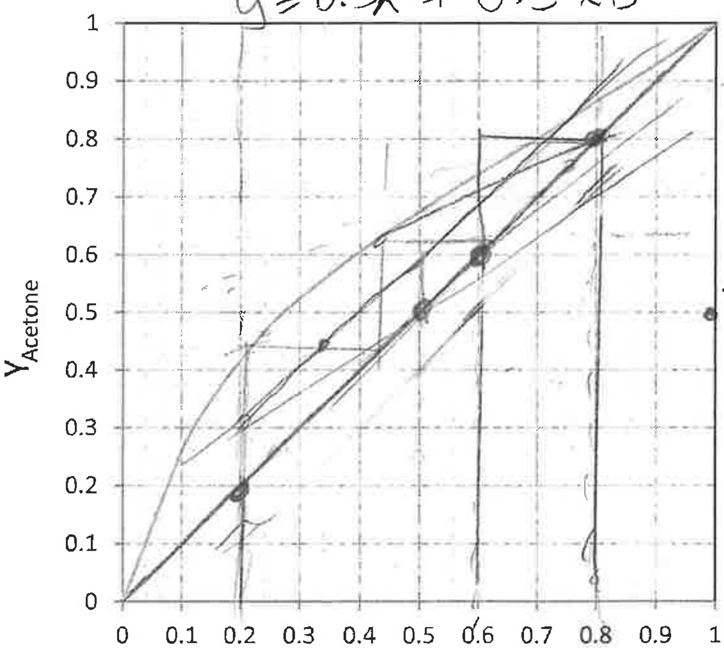
10.0 kmoles of a feed that is 0.80 mole fraction acetone and 0.20 mole fraction ethanol is batch distilled in a system with a still pot (an equilibrium contact), a column that acts as one equilibrium stage (total two equilibrium contacts), a total condenser, and reflux to the column. p = 1.0 atm, the reflux ratio is L/D = 1.0, CMO is valid, and $x_{W,final}$ = 0.20.

1) (2pts) Find the slope of operation equation, $y = (L/V)x + (1-L/V)x_D$ 2/2
2) (4pts) Complete the table below using the VLE graph (draw OP lines and mark your x_w values on the graph), and mark points for area calculation.

1

	x_W	x_D	$x_D - x_W$	$1/(x_D - x_W)$
$x_{W,initial}$	0.80			
$x_{W,middle}$	0.60			
$x_{W,final}$	0.20			

$y = 0.5x + 0.5x_D$



1) $\frac{L}{V} = \frac{L/D}{1 + \frac{L}{D}}$

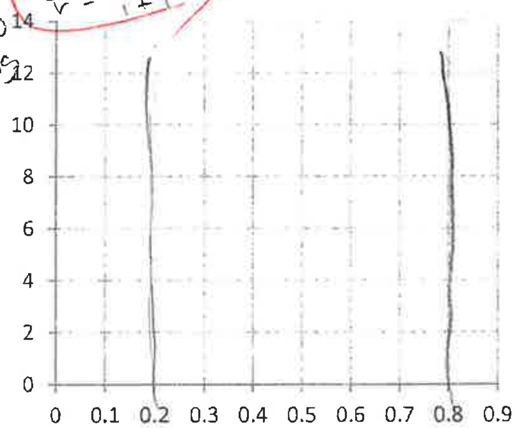
$\frac{L}{V} = \frac{1}{1+1}$

$x_F = 0.80$

$x_f = 0.20$

x_D is what's moving

this helps us find W_f & W_F



$W_f = F e^{-(area)}$

→ after you find area get W_f

1

3) (4pts) Find W_{final}, D_{total}, and $x_{D,avg}$.

$\text{Area} = \frac{0.80 - 0.20}{6} = \left[\frac{1}{x_D - x_w} + 4\left(\frac{1}{x_D - x_w}\right)\right.$ $\frac{x_F - x_f}{6} = \left[W_f + 4(\quad) + W_F\right]$

$W_f + \frac{1}{x_D - x_w}\Big|_{W_F}$

$0.1 [\quad]$

$D_{tot} = F - W_f$

Simpson's rule: $Area = \frac{x_{Feed} - x_{final}}{6}\left[\frac{1}{x_D - x_w}\bigg|_{x_{final}} + \frac{4}{x_D - x_w}\bigg|_{x_{middle}} + \frac{1}{x_D - x_w}\bigg|_{x_{Feed}}\right]$

Rayleigh equation: $W_{final} = F\,exp\left(-\int_{x_{final}}^{x_{Feed}} \frac{dx_w}{(x_D - x_w)}\right) = F\,exp(-Area)$

$x_{D,avg} = (F * x_F - W_{final} * x_{final})/D_{total}$